

Lecture 1.9 – Factors, Special Values, and Class Coercion

Specific Learning Objectives:

1.1.9 – Understand the data classes of R.

1.1.13 – List the special values of R.

1.1.14 – Create ordered and unordered factors.

1.1.15 – Learn to coerce data from one class to another.

So far we've learned...



- A lot of different types of data and ways to organize them!
 - Types of data: numeric, integer, complex, logical, character.
 - Useful ways of organizing them (classes): matrix, data frame, list
- Today we'll learn an additional type of data/specialized class (factor).
- We'll also learn about special numbers which don't quite fit into other types.
- Finally, we'll learn how to turn one type of data or class into another (called coercion).

Factors

- Factor is a class of data that is special to R.
- Factors convert data into categorical data.

```
> colors <- c("white","blue","red", "blue", "red", "red")
> color.fac <- factor(colors)
> color.fac
[1] white blue  red   blue  red   red
Levels: blue red white
```

1 2 3

Values	
color.fac	Factor w/ 3 levels "blue","red","white": 3 ...
colors	chr [1:6] "white" "blue" "red" "blue" "red"...

- Each category is represented by a name and encoded with integers.

```
> as.numeric(color.fac)
[1] 3 1 2 1 2 2
```



**This saves a lot of
memory with large
data sets!**

Creating Factors

- Factor creation is easy.

- Create factored data *de novo* with `factor()`:

```
> pets <- factor(c("dog", "cat", "squirrel", "dog", "cat", "dog", "dog", "bear", "cat",  
+                 "fish"))
```

**Text, numbers, etc must
be exact matches for
`factor()` to work
properly!**

- ▶ `factor()` takes a vector as its first argument.
- ▶ `factor()` takes `levels` as an argument if you want to assign specific levels.

- You can also use an existing vector with `factor()`:

```
> colors <- c("white","blue","red", "blue", "red", "red")  
> color.fac <- factor(colors)
```

Note: factors are only generated in vector form!

Creating Factors

- Factors can be either *unordered* or *ordered*.
 - Unordered factors: blue, red, white
 - Ordered factors: good, better, best
- Create ordered factors also using **factor()**: use argument **ordered=TRUE** and add **levels** to specify order.

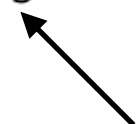
```
> pet.quality <- c("great", "good", "ok", "great", "good", "great", "great", "poor", "good",  
+                 "ok")  
> pet.quality <- factor(pet.quality, levels=c("poor", "ok", "good", "great"), ordered=T)
```

Here we are overwriting
pet.quality with the new
factored vector!



```
> pet.quality  
[1] great good ok great good great great poor good ok  
Levels: poor < ok < good < great
```

Here is the order of your levels!



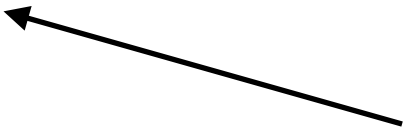
Factor Info

- `levels()` will return the levels of your factor

```
> levels(pets)
[1] "bear"      "cat"       "dog"       "fish"      "squirrel"
```

```
> levels(pets)[1]
[1] "bear"
```

You can select a single level by specifying a number in square brackets after the `levels()` function.



```
> levels(pet.quality)
[1] "poor"  "ok"    "good"  "great"
```

← On ordered data, `levels()` will report in their order.

- You can access the elements of factored vector the same as other vectors! As are:
 - Replacing elements
 - Subsetting
 - Basically they are just like regular vectors

Check Your Understanding

Create a data frame that contains:

- a column of class character describing something in your current surroundings**
- a column of class factor describing the color of those items**

When data doesn't quite fit (or isn't there)...

- There are sometimes numbers (or non-numbers) that pop up in data sets and calculations. They should be treated specially by R, so they have special values.
 - Infinity
 - Not-a-numbers
 - Data not available
 - NULL values

Infinity

- Infinity occurs when *R* is forced to limit the size of a number. When it is too large, it is considered by *R* to be infinite (mathematical symbol: ∞).
- Infinity is **Inf** in *R*
- Infinity can be positive (**Inf**) or negative (**-Inf**)
- Once an element is infinite, all operations on that element will result in another infinite number.
- If you divide a number by **Inf**, the result is 0 (zero).
- You can check an *R* object for infinities using **is.infinite()** and/or **is.finite()** which will return a logical for each element.

Not Available

- When you have data that is unknown, uncollected, or otherwise “not available”, you want a special value to indicate this in a data set.
 - R represents not-available data with the value **NA**
 - Elements having **NA** will not accept any manipulation other than replacing the element! R will completely ignore this element.
 - **NA** serves as a useful placeholder
 - You can check for the presence of **NA** by using `is.na()` which will return a logical for each element in an R object.
 - You can remove any observations with an **NA** by using `na.omit()`.

Not-a-number

- When you perform a calculation that results in a value that is not a number, such as **Inf/Inf** or **0/0**.
 - R represents not-a-number elements with the value **NaN**
 - Elements having **NaN** will not accept any manipulation other than replacing the element! R will completely ignore this element.
 - **NaN** is functionally the same as **NA** but holds a different meaning.
 - You can check for the presence of **NaN** by using **is.nan()** which will return a logical for each element in an R object.

NULL

- **NULL** indicates an empty entity (value, vector, etc).

- **NULL** is different than **NA**:

```
> c(NA,NA,NA)  
[1] NA NA NA
```

```
> c(NULL,NULL,NULL)  
NULL
```

- ▶ In this example, **NA** represents three missing values, so they exist or could be known, but are not recorded. This is reflected in the output, recording **three spots of NA!**
 - ▶ However, **NULL** in three empty places (or emptiness three times) is interpreted as a **single empty entity.**
- Check for **NULL** using `is.null()`

Special values in practical use

– When to use each:

- **Inf** - do not enter
- **NaN** - do not enter
- **NA** - use this when data is missing
- **NULL** - Use this as a place holder only for entire objects

– What each practically means:

- **Inf** / **NULL** / **NaN** - a calculation did something unexpected (divided by zero), you should go back and check to see what went wrong
- **NA** - data is missing, will screw up some functions.

Options:

- ▶ Use **na.rm** argument

```
> avec <- c(1, 2, 3, NA, 5)
> mean(avec)
[1] NA
> mean(avec, na.rm = TRUE)
[1] 2.75
```

- ▶ Use **na.omit()** to clean data frames, will remove observations with any **NA** (in other words, rows!)

Check Your Understanding

What would you do in the following scenarios?

- a) You have a data set in a data frame. It is very large, and you want to know whether or not it has missing data values. You'd also like to remove those values so that you only have a set of complete observations.

- b) You have a data frame in which you use your own function with vectorized calculations. When you look at the output, it is `NULL`. What has happened? What should you do?

Class and Data-type Coercion

- Sometimes you will need data types or classes to be converted into another class or type in order to perform some action. Converting one type or class of data into another is called **coercion**.
 - Coercion can happen automatically, such as logical to numeric with `sum()` or integer to numeric with many different functions.
 - Sometimes you will have to coerce data manually, this is most often accomplished with `as.` functions:
 - ▶ `as.character()`
 - ▶ `as.numeric()`
 - ▶ `as.integer()`
 - ▶ `as.factor()`
 - ▶ `as.logical()`
 - ▶ `as.data.frame()`

Class and Data-type Coercion

- Dates and times are a special circle of h*ck in R, but it is useful to know how to use a couple functions.
 - ▶ **POSIX** is a standardized date format that is accepted across several languages. Many plotting commands will use this format, so it makes plotting with dates much nicer.
 - ▶ **as.POSIXct()** converts a character date and time in a format (2022/07/12 10:00:30) to a POSIX object representing the number of seconds since Jan 1, 1970.
 - ▶ **as.POSIXlt()** converts a character date and time in a format (2022/07/12 10:00:30) to a list that stores date, hours, minutes, etc as elements.

Check Your Understanding

Which command can accomplish the following coercions:

- a) a logical vector into a numeric vector**
- b) a numeric vector into a character vector**
- c) two vectors into a data frame**
- d) a list into a vector**

Action Items

- 1. Complete Assignments 1.11 and 1.12.**
- 2. Prepare for your first Exam!**